74+10 = 84

ST10440943 - SAHEEL RAMONE BHUGWANDEEN

INSY7311 ASSIGNMENT 1

UniLife Manager Mobile Application Report

10

1. Introduction

University students often face challenges when trying to balance academic responsibilities and personal life. Managing assignments, deadlines, finances, events, and personal goals can become overwhelming when these responsibilities are organised across multiple tools or applications. To address this challenge, a mobile application called **UniLife Manager** is proposed. UniLife Manager is designed to assist students in organising both their academic and personal activities within a single, easy-to-use mobile platform.

The application provides several features that support student productivity and organisation. These include an assignment and deadline tracker, a personal to-do list, a budget and expense tracker, event reminders, and a personal goal tracker. By integrating these functions into one application, UniLife Manager helps students organise their tasks efficiently and manage their daily responsibilities.

When designing mobile applications, it is important to consider both **usability** and **user experience (UX)**. Usability refers to how effectively, efficiently, and satisfactorily users can achieve their goals when using a system (Nielsen, 1994). For example, in UniLife Manager, clearly labelled buttons such as “Add Assignment” or “Add Task” allow students to perform actions quickly without confusion.

User experience (UX) focuses on the overall feelings and perceptions users have when interacting with a system. UX design aims to create meaningful and satisfying interactions by ensuring that systems are intuitive, efficient, and enjoyable to use (UXMatters, 2020). In UniLife Manager, a clean dashboard layout, clear navigation, and helpful feedback messages improve the overall experience of the user.

Therefore, while usability ensures that the application is easy to use, user experience focuses on how enjoyable and satisfying the interaction with the system is. Both concepts are important for ensuring that UniLife Manager effectively supports students in managing their academic and personal activities.

2. User-Centred Design (UCD)

8

User-Centred Design (UCD) is an approach to system design that focuses on the needs, behaviours, and goals of users throughout the development process. This design philosophy ensures that the system is built around how users interact with the product rather than forcing users to adapt to the system (Interaction Design Foundation, 2024).

In the development of UniLife Manager, the primary users are university students who need a simple and efficient way to manage both academic and personal responsibilities. By applying a user-centred design approach, the application prioritises features that directly support student needs, such as quick access to assignments, reminders for deadlines, and tools for organising tasks.

User-centred design is important because it improves usability, effectiveness, and user satisfaction. Systems designed around user needs are easier to understand and require less effort to learn. UCD also encourages continuous evaluation and testing with users during the design process to identify usability problems and improve the interface (Shneiderman et al., 2016).

For UniLife Manager, applying UCD ensures that the application remains intuitive, practical, and relevant to students' daily activities. By focusing on the users' requirements and behaviours, the application can better support students in managing their academic workload and personal organisation.

3. Data Gathering Plan

8 Understanding the needs of users is an essential step in the design of any application. User research helps designers identify user behaviours, motivations, and challenges, which can then inform the development of effective solutions (Interaction Design Foundation, 2024). For UniLife Manager, two data gathering methods are recommended: **surveys** and **interviews**.

3.1 Surveys

Surveys are a structured method of collecting information from a large group of participants through questionnaires. Surveys are useful for identifying patterns in user behaviour and understanding the common challenges that users experience.

For UniLife Manager, surveys could be distributed to university students to gather information about how they currently organise their assignments, track their tasks, and manage their personal responsibilities. Survey questions may ask students whether they use multiple apps for organisation and what features they would prefer in a student management application.

Surveys are beneficial because they allow researchers to gather data from many participants quickly, making it easier to identify common trends and user needs (Nielsen, 1994).

3.2 Interviews

Interviews involve direct conversations with participants and allow researchers to collect detailed information about user experiences. Unlike surveys, interviews allow participants to explain their challenges and behaviours in greater depth.

For UniLife Manager, interviews with students could reveal insights into how they manage their academic workload, what problems they encounter when organising tasks, and which tools they currently use. These insights help designers better understand user needs and expectations (Preece, Rogers, & Sharp, 2015).

Using both surveys and interviews provides a combination of quantitative and qualitative data, allowing designers to gain a more comprehensive understanding of user requirements.

4. Low-Fidelity Prototype

A low-fidelity prototype is a simple visual representation of an application that demonstrates layout and functionality without focusing on detailed graphics. Low-fidelity prototypes are commonly used during the early stages of development to assess ideas and evaluate design principles before building the final system (Norman, 2013).

The UniLife Manager prototype demonstrates several design principles proposed by usability and human-computer interaction researchers.

4.1 Design Principle 1 – Visibility

8 Visibility refers to ensuring that important functions are clearly visible to users so that they can easily identify what actions are possible (Norman, 2013).

In the UniLife Manager application, the home dashboard displays key features such as Assignments, Tasks, Budget, and Events using clearly labelled buttons.

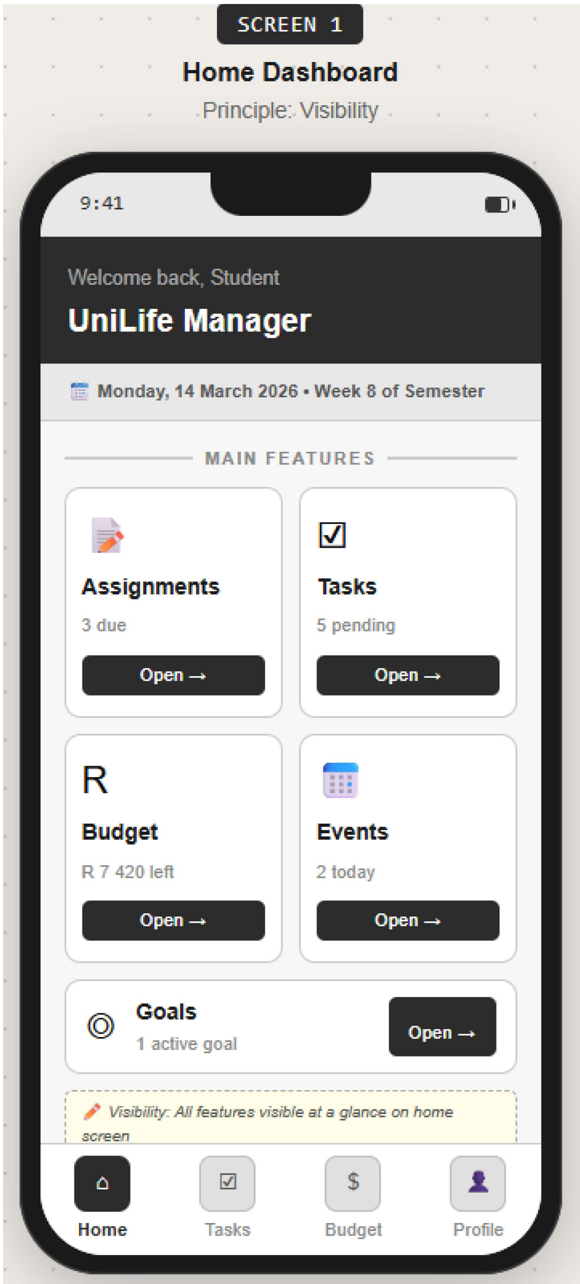

Figure 1: Home Dashboard demonstrating the Visibility principle.

This design allows students to immediately see the main functions of the application when they open it.

4.2 Design Principle 2 – Constraints

8 Constraints are design elements that limit user actions to prevent mistakes and guide users toward correct interactions (Norman, 2013).

In the UniLife Manager application, the “Add Assignment” screen includes structured fields such as assignment title, course name, and due date. The due date is selected using a calendar picker, which prevents incorrect formatting.

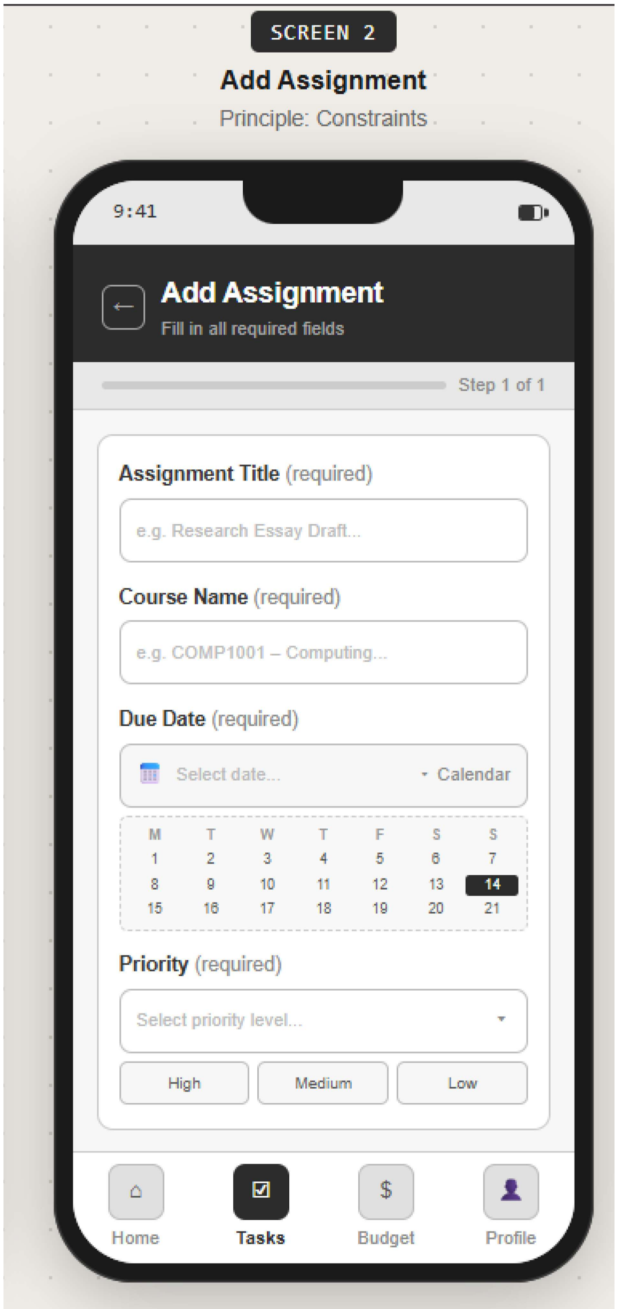

Figure 2: Add Assignment screen demonstrating Constraints.

These constraints help ensure that users enter valid information.

4.3 Design Principle 3 – Feedback

Feedback refers to the system's response to user actions. Providing immediate
8 feedback helps users understand whether their action was successful (Norman,
2013).

In UniLife Manager, after a user adds a new assignment or task, a confirmation message appears informing the user that the task has been saved successfully.

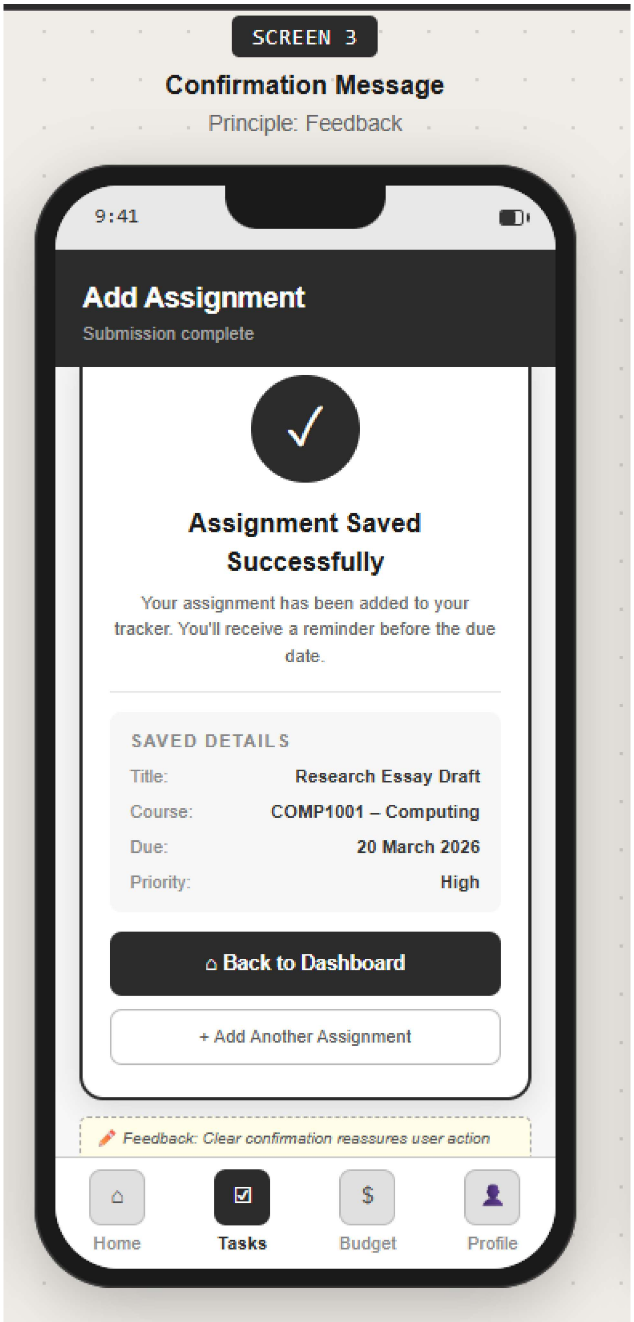

Figure 3: Confirmation message showing system Feedback.

This feedback reassures users that their information has been recorded.

4.4 Design Principle 4 – Error Prevention

8 Error prevention is a principle that focuses on designing systems that help users avoid making mistakes (Shneiderman et al., 2016).

In UniLife Manager, if a user attempts to save an assignment without completing all required fields, the system displays an error message prompting the user to complete the missing information.

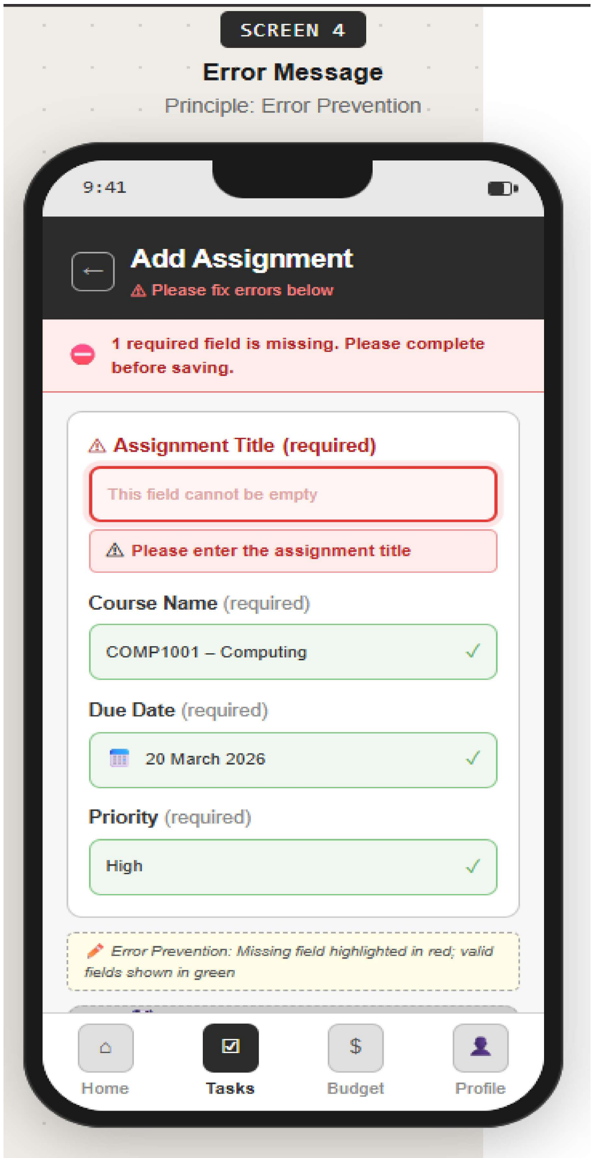

Figure 4: Error message demonstrating Error Prevention.

This feature prevents incomplete information from being stored and guides the user to correct the error.

4.5 Design Principle 5 – Consistency

8 Consistency ensures that similar elements and actions behave the same throughout the application. Consistent design patterns reduce confusion and help users learn the system more quickly (Shneiderman et al., 2016).

UniLife Manager maintains consistency by using the same navigation layout across all screens.

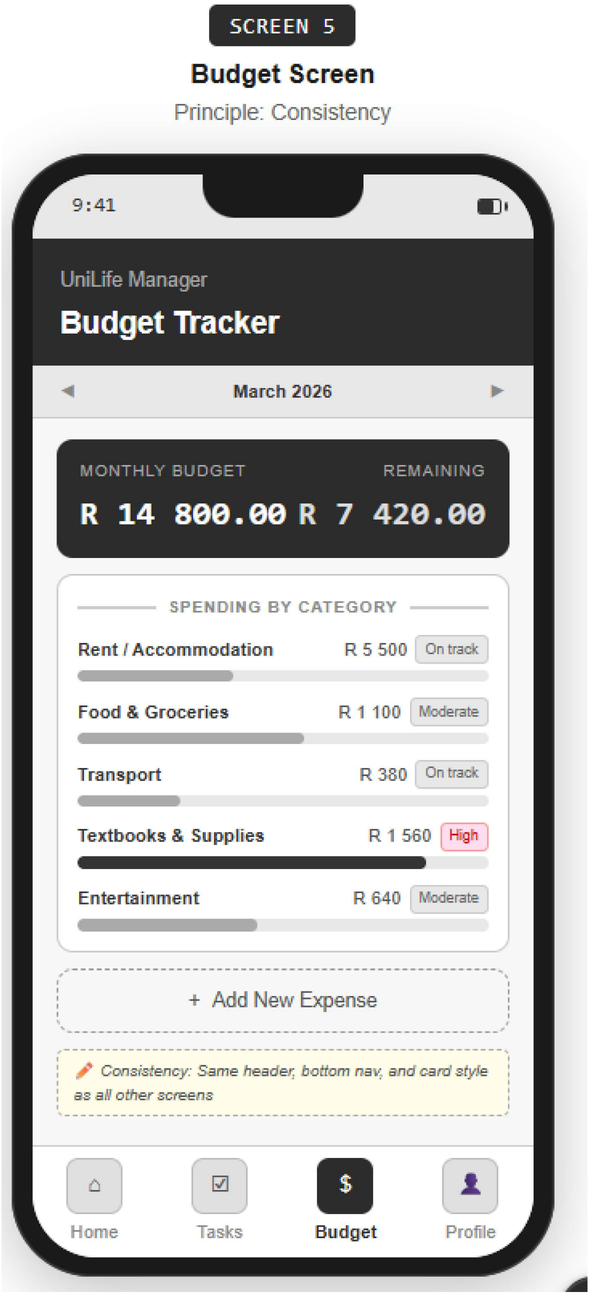

Figure 5: Bottom navigation showing Consistency.

Consistent design makes it easier for users to navigate the application and locate features.

5. Navigation and Menus

8 Navigation is an important aspect of mobile application usability. Effective navigation allows users to move between different sections of an application easily and complete tasks efficiently (Shneiderman et al., 2016).

UniLife Manager uses a **bottom navigation menu**, a common interaction design pattern in mobile applications. The navigation bar includes icons and labels for the main sections of the application: Home, Tasks, Budget, Events, and

Profile. This structure allows users to quickly access key features without navigating through multiple screens.

Clear navigation reduces cognitive effort and improves usability (Nielsen, 1994). Bottom navigation menus are recommended in mobile interface design because they allow users to easily access key features using natural thumb movements (UXMatters, 2020).

6. References

Interaction Design Foundation. (2024). *User-centered design*. Retrieved from https://www.interaction-design.org/literature/topics/user-centered-design

Interaction Design Foundation. (2024). *User research – methods and best practices*. Retrieved from https://www.interaction-design.org/literature/topics/user-research

Nielsen, J. (1994). *Usability engineering*. Academic Press.

Norman, D. A. (2013). *The design of everyday things* (Revised and expanded edition). Basic Books.

Preece, J., Rogers, Y., & Sharp, H. (2015). *Interaction design: Beyond human-computer interaction* (4th ed.). Wiley.

Shneiderman, B., Plaisant, C., Cohen, M., Jacobs, S., Elmqvist, N., & Diakopoulos, N. (2016). *Designing the user interface: Strategies for effective human-computer interaction* (6th ed.). Pearson.

UXMatters. (2020). *Universal design principles and heuristic guidelines*. Retrieved from https://www.uxmatters.com